

Trust-Based Cross-Domain Batch Authentication Protocol for UAV Networks

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Abstract—Existing cross-domain authentication protocols for multiple unmanned aerial vehicles (UAVs) suffer from low efficiency. Batch authentication serves as an effective approach to improve authentication efficiency. However, in cross-domain environments, this method is vulnerable to pollution attacks, which can result in authentication failures. To address these issues, this paper proposes an efficient cross-domain batch authentication scheme for UAV networks. Specifically, a cross-domain key update mechanism based on physical unclonable functions (PUFs) is designed to resist external pollution attacks. A blockchain-based cross-domain trust management mechanism is proposed to prevent internal pollution attacks. Furthermore, a software-defined networking (SDN)-based pre-negotiation mechanism is developed to eliminate the additional overhead caused by trust detection during cross-domain real-time authentication. Security analysis shows that the proposed scheme achieves multiple security properties. Experimental results indicate that our scheme reduces computational overhead by at least 15.27% compared to related schemes, while parameter optimization can significantly enhance the robustness of the trust model.

Index Terms—unmanned aerial vehicle network, trust, cross-domain batch authentication, blockchain, physical unclonable function

I. INTRODUCTION

Unmanned aerial vehicles (UAVs) play a crucial role in the low-altitude economy. Their flexible deployment capabilities and advanced sensing capabilities offer essential support across various fields [1]. As missions become more complex, UAV systems need to evolve from working within isolated areas to facilitating cooperation between different domains [2]. This change necessitates collaboration among entities from separate trust areas to work on shared tasks and resources. Such teamwork not only removes administrative obstacles but also enhances efficient and dynamic resource management [3].

The problem of unauthorized access to UAVs brings about considerable security issues, such as possible privacy violations and harmful cyber-attacks [4]. Authentication is the main method used to confirm the legitimacy of participants in these situations [5]. However, UAV networks encounter significant obstacles in cross-domain authentication. The initial challenge stems from the varied authentication protocols used across different trust domains. This diversity in protocols hinders local ground stations (GS) from directly verifying UAVs registered in external domains [6]. The second challenge is the risk of malicious actions from properly registered UAVs within the target domains.

Although existing cross-domain authentication schemes focus primarily on individual terminal devices, practical scenarios often require GS to handle concurrent authentication requests from multiple UAVs across trust domains. Batch authentication presents an effective solution to enhance authentication efficiency in such environments [7]. This approach enables simultaneous verification of identity for multiple UAVs using signature aggregation techniques, where GS can consolidate multiple individual signatures into a single verifiable object [8]. However, these batch authentication mechanisms remain vulnerable to pollution attacks, in which adversaries submit invalid signatures that cause a complete batch verification failure and service denial. Typical pollution attacks in cross-domain batch authentication are categorized into two distinct types.

External pollution attacks: Attackers exploit vulnerabilities in inter-domain trust relationships. They acquire cross-domain authentication key materials through external methods such as physical capture. Subsequently, they impersonate legitimate UAVs to disrupt batch authentication processes.

Internal pollution attacks: Attackers maintain a legitimate status in their registration domain, but carry out malicious activities on the target domains. This attack variant demonstrates high concealment capabilities. It induces cross-domain batch authentication failures by periodically transmitting invalid signatures.

This paper proposes an efficient cross-domain batch authentication scheme addressing these security challenges. To counter external pollution attacks, we develop a private key update mechanism based on physical unclonable functions (PUFs). This mechanism ensures secure cross-domain private key transmission while providing forward security for the protocol. For internal pollution attacks, we establish a blockchain-based trust management mechanism that continuously monitors and isolates UAVs exhibiting malicious behavior, effectively preventing them from disrupting batch authentication through persistent transmission of invalid messages. Furthermore, we develop a Software-Defined Networking (SDN)-based pre-negotiation mechanism which leverages the SDN controller's global view and programmability to proactively orchestrate authentication handovers. By predicting UAV mobility and pre-validating credentials with adjacent domains, this mechanism shifts the computational burden of trust verification from the real-time authentication phase to a

preparatory phase, thereby eliminating the associated overhead during critical handovers. Security analysis confirms the robust security properties of our scheme, while performance comparisons with recent schemes demonstrate significantly reduced computational overhead.

II. SYSTEM MODEL

A. System Model

As illustrated in Fig.1, the proposed authentication framework comprises four fundamental components that collaboratively establish secure cross-domain communication for UAV networks: the SDN controller, GS, UAV nodes, and a dual-layer blockchain infrastructure. The functional roles and security relationships among these entities are detailed as follows:

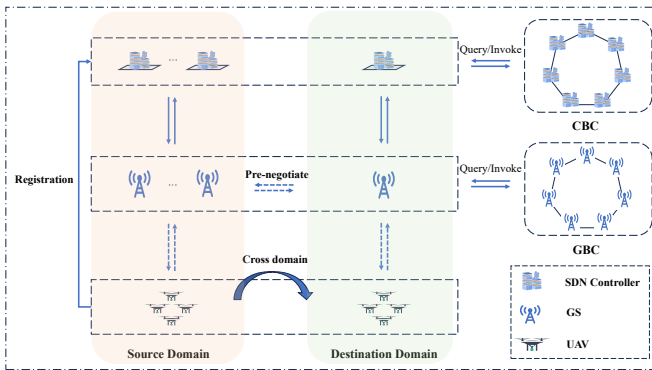


Fig. 1. System model.

SDN controller: SDN controller: This component operates as a trusted administrative entity and the core driver of the pre-negotiation mechanism within the trust domain. It maintains superior computational and storage resources. Beyond overseeing UAV registration and distributing credentials, the controller proactively triggers cross-domain pre-negotiation for UAVs based on real-time flight trajectory predictions. This involves pre-verifying trust credentials and coordinating with neighboring domains via the blockchain, ensuring that authentication materials are in place before a UAV initiates a cross-domain request. This SDN-driven approach eliminates the overhead of real-time trust detection during actual handovers. Additionally, the controller continuously evaluates UAV trust scores through CBC log analysis and administers the revocation process for compromised identities.

GS: Serving as semi-trusted infrastructure components, GS units provide intermediate computational and storage performance. They implement single-instance and batch authentication procedures for local and cross-domain UAVs. Upon completion of the authentication, the GS devices transfer verification records and UAV flight information to their domain SDN controller. This data is then relayed to the CBC network. Throughout pre-negotiation cycles, GS components evaluate UAV suitability for batch authentication. These evaluations employ trust indicators from the SDN controller to detect

potentially malicious actors in cross-domain verification scenarios.

UAV: These devices constitute vulnerable network endpoints that are subject to physical compromise. They function under strict computational and memory constraints. Consequently, all UAVs must undergo an initial registration with their designated domain SDN controller. Secure inter-domain communication with GSs is permitted only after successful authentication.

Blockchain: The architecture employs a dual-layer blockchain framework that integrates the CBC and GBC networks. CBC operates as a consortium blockchain connecting regional SDN controllers. Its primary functions encompass trust assessment and valid identity revocation for malicious entities. Through smart contract implementation, CBC enforces strict access control for authorized SDN controllers. Its distributed ledger maintains pseudonym UAV identifiers, identity restoration parameters, authentication transcripts, and historical trust measurements. GBC constitutes a consortium blockchain that unifies all GSs in the system. Its principal operation facilitates cross-domain authentication mechanisms. Similarly, governed by smart contracts, GBC imposes access restrictions on validated GS members. The immutable ledger stores encrypted temporary UAV identifiers, pseudonymous identities, session challenges, and ephemeral public keys.

B. Trust Management Mechanism

The trust management framework establishes a dynamic evaluation system that ensures reliable identification of malicious UAVs. This mechanism operates through four interconnected phases that collectively form a comprehensive trust lifecycle management system.

Blockchain Preparation: Upon completing cross-domain authentication, the local GS forwards the authentication logs to the domain SDN controller. The controller uploads these logs to IPFS, generating a unique hash value CID. The CID, PID, and corresponding timestamp are then stored in the CBC through smart contracts. The CBC functions as a distributed ledger that records authentication data from all participating nodes, ensuring tamper-resistant storage.

Historical Trust Computation: To reduce the computational load on the GSs, the SDN controllers assume responsibility for calculating the trust values. When an operational cycle is completed, each regional SDN controller periodically retrieves authentication logs for all IPFS UAVs. The initial step involves verifying the consistency between the hash on the chain and the IPFS data. After successful verification, the controller calculates the historical trust value T_{old} for each UAV. The final step involves uploading T_{old} to the CBC via a smart contract.

Trust Evaluation: During the pre-negotiation phase, the target domain's SDN controller fetches the corresponding file CIDs and historical trust values T_{old} from the CBC. It then obtains the UAV's authentication logs for the current cycle

from IPFS. If T_{old} exceeds a predefined threshold, the updated trust value is computed as:

$$T_{new} = \lambda \cdot T_{old} + (1 - \lambda) \cdot \left[\frac{S_{batch} + 1}{S_{batch} + F_{batch} + 2} - \beta \cdot \max(0, F_{batch} - k) \right] \quad (1)$$

Here, λ is the decay factor that reduces the influence of historical data. S_{batch} counts successful batch authentications in the current cycle. F_{batch} counts failed batch authentications. β is the penalty factor for failures.

Since all UAVs might experience increased batch authentication failures due to malicious actors or environmental issues, a robustness factor k is incorporated. The appropriate configuration of k helps mitigate the impact of malicious users on legitimate UAVs. When $F_{batch} > k$, the trust value incurs an additional penalty, accelerating its decline.

Trust Recovery: UAVs may be misclassified as malicious due to signal interference or hardware failure during operation. The proposed recovery mechanism corrects these errors through continuous monitoring and re-evaluation. If a UAV's T_{old} does not exceed the threshold, it is excluded from batch authentication for that cycle. After the cycle ends, the SDN controller updates the UAV's trust value based on its individual authentication records:

$$T_{new} = \min \left(\left[\lambda \cdot T_{old} + (1 - \lambda) \cdot \gamma \cdot \left(\frac{S_{single} + \alpha}{S_{single} + F_{single} + \alpha} \right) \right], T_{thresh} + \sigma \right) \quad (2)$$

Here, γ is the recovery factor. S_{single} counts successful individual authentications in the current cycle. F_{single} counts failed individual authentications. σ is a minimal constant value.

Trust Decision: The trust threshold T_{thresh} is set using dynamic or static methods. A UAV is deemed trustworthy and allowed to join batch authentication between domains if its trust value exceeds T_{thresh} . Otherwise, it is classified as malicious and is barred from batch authentication. UAVs identified as malicious must execute a trust recovery algorithm in subsequent cycles to restore their trust values.

III. PROPOSED SCHEME

A. Registration Phase

UAVs are required to complete an identity registration in their home domain. The registration process involves the following steps:

Step 1: The UAV sends a registration request to its domain's SDN controller, which contains the UAV's real identity, denoted as ID_i .

Step 2: Upon receiving the request, the SDN controller checks the uniqueness of ID_i . If it is unique, the controller generates a random number and calculates the necessary

cryptographic parameters. It then sends the challenge value C_i and the pseudo-identity PID_i back to the UAV.

Step 3: The UAV uses its PUF to process the challenge C_i and produce the response R_i . It then creates a public-private key pair (PK_i, PR_i) and a temporary identity TID_i . These, PK_i and TID_i , are sent back to the SDN controller.

Step 4: The SDN controller then stores (C_i, PR_i, PK_i) along with the hash value $H(TID_i)$ securely in the GBC.

B. Pre-negotiation Phase

In the source domain, the GS continuously monitors the UAV parameters, such as signal strength, location data, and flight attributes, during mission operations. The SDN domain controller utilizes this data to predict potential target GSs. When there is a need for cross-domain authentication, the pre-negotiation protocol is initiated following these steps:

Step 1: The source GS sends the TID_i of the UAV to the target GS. The target GS then queries the GBC to obtain the corresponding PID_i and forwards this data to the domain SDN controller.

Step 2: The SDN controller retrieves the UAV's authentication credentials and its historical trust value T_{old} from the CBC. If T_{old} is below a specified threshold, the batch authentication flag η is set to 0. Otherwise, the controller runs the trust evaluation algorithm to compute a new trust value, setting the batch authentication flag η to 1.

Step 3: The target GS gathers batch authentication flags from all UAVs and conducts grouping operations. UAVs with $\eta = 1$ are grouped for batch authentication, while those with $\eta = 0$ are placed in the single authentication group for individual verification.

C. Cross-Domain Batch Authentication Process

When a UAV, denoted as UAV_i , enters a new trust domain, mutual authentication with the target domain's ground station GS_j is necessary. The detailed procedure consists of the following steps:

Step 1: The UAV initiates a cross-domain authentication request by sending its TID_i to the target GS. The GS then queries the GBC using $H(TID_i)$ to retrieve $\{PID_i, C_i, PK_i\}$. It generates a random number u_i , computes $U_i = u_i \cdot P$, and sends back the message $\{C_i, U_i\}$ to the UAV.

Step 2: Upon receipt, the UAV randomly generates C_{i+1}, v_i , and t_1 . It then determines the PUF response R_i using C_i as input. The temporary private key is calculated as $PR_i = H(R_i)$. The UAV computes $V_i = v_i \cdot P$ and $h_1 = H(PID_i \parallel ID_j \parallel C_{i+1} \parallel PK_{i+1} \parallel V_i \parallel A_i \parallel t_1)$. It then creates the signature $\tau_i = v_i + h_1 \cdot PR_i$. The UAV updates its PUF response to $R_{i+1} = f_{PUF}(C_{i+1})$ and generates the next temporary private key $PR_{i+1} = H(R_{i+1})$, with the corresponding public key $PK_{i+1} = PR_{i+1} \cdot P$. It calculates the shared parameter $A_i = U_i \cdot v_i$ and the encrypted message $M_1 = (C_{i+1} \parallel PK_{i+1}) \oplus H(A_i)$. Finally, the UAV sends $\{\tau_i, V_i, M_1, t_1\}$ to the target GS.

Step 3: The target GS first checks the freshness of the timestamp t_1 . If the timestamp is invalid, the authentication process

is halted. If valid, it recalculates $h_1 = H(PID_i \parallel ID_j \parallel C_i \parallel V_i \parallel A_i^* \parallel t_2)$ and verifies the equation $\tau_i.P = V_i + h_1.PK_i$. If this verification fails, the authentication is aborted. For simultaneous authentication requests from multiple UAVs in different trust domains, the GS employs aggregated signature verification:

$$\sum_{i=1}^n (\tau_i).P = \sum_{i=1}^n V_i + \sum_{i=1}^n h_1.PK_i$$

Following successful verification, the GS generates the timestamp t_2 and computes $A_i^* = u_i.V_i$. It recovers update information through $(C_{i+1}^* \parallel PK_{i+1}^*) = M_1 \oplus H(A_i^*)$ and generates a new pseudonym $TID_i^{new} = PRNG(TID_i)$. The session key is calculated as: $SK_{j-i} = H(TID_i^{new} \parallel ID_j \parallel C_{i+1}^* \parallel PK_{i+1}^* \parallel A_i^*)$. The GS generates the authentication message $M_2 = H(SK_{j-i} \parallel t_2)$, uploads $\{H(TID_i^{new}), PID_i, C_{i+1}^*, PK_{i+1}^*\}$ to the GBC, and transmits the response $\{M_2, t_2\}$ to the UAV.

Step 4: The UAV validates the freshness timestamp t_2 . Invalid timestamps terminate the process. Otherwise, it computes $TID_i^{new} = PRNG(TID_i)$ and derives the session key: $SK_{i-j} = H(TID_i^{new} \parallel ID_j \parallel C_{i+1} \parallel PK_{i+1} \parallel A_i)$. The UAV verifies if the calculated $M_2^* = M_2$ matches the received value. Successful verification confirms mutual authentication. Otherwise, authentication fails.

IV. SECURITY ANALYSIS

This section provides a comprehensive security analysis of the proposed authentication scheme, demonstrating its resilience against various attack vectors while fulfilling critical security requirements. The following aspects are systematically examined:

Cross-Domain Batch Authentication. The system supports simultaneous verification of multiple UAVs originating from identical or distinct trust domains. The GS performs batch verification through the aggregate signature operation.

Anonymity and Unlinkability. Our scheme implements an anonymous identification framework through pseudonym substitution. The real identity ID_i is replaced with a pseudonymous identity PID_i for blockchain storage. During cross-domain authentication, the temporary identity TID_i provides additional protection. This layered approach effectively preserves identity anonymity throughout system operations.

Impersonation Attacks. Successful UAV impersonation requires forgery of authentication $\{\tau_i, V_i, M_1, t_1\}$. The signature component τ_i contains PUF-generated ephemeral credentials that are fundamentally unforgeable due to physical unclonability. Similarly, GS impersonation attempts require fabrication of $\{M_2, t_2\}$. However, M_2 incorporates the session key SK_{j-i} , where security derives from the parameter protected by the ECC.

Replay Attacks. The timestamp mechanisms embedded in the authentication messages M_1 and M_2 provide replay protection. The receiving entities validate the temporal freshness before processing the authentication materials. This timestamp

verification ensures the rejection of replayed messages, eliminating replay attack vulnerabilities.

Physical Capture Attacks. Physical compromise of UAV nodes produces no persistent key material. The system never stores PUF-generated private keys in non-volatile memory. The uniqueness of PUF and physical unclonability prevent duplication of credentials through circuit analysis. Therefore, key extraction through power analysis becomes computationally infeasible.

Pollution Attacks. The scheme addresses two variants of the pollution attack. For external pollution attacks, adversaries cannot obtain the required private keys, since the credentials are ephemeral and PUF-derived. Internal pollution attacks from malicious legitimate entities are prevented through trust management mechanisms. Dynamic screening identifies and isolates potentially malicious UAVs prior to batch authentication, preventing pollution through invalid signature submission.

V. PERFORMANCE EVALUATION

In this section, we perform a comparative analysis with existing mobile IoT handover schemes and validate the effectiveness of our trust model. To emulate realistic heterogeneous computing environments, cryptographic operations are implemented on different hardware platforms. The infrastructure side employs a laptop with an Intel Core Ultra 9 processor and 16 GB RAM. The device side utilizes a Raspberry Pi 5 featuring a Cortex A76 processor and 8 GB RAM. We measure the time consumption for cryptographic operations using the MIRACL library. Specifically, the ECC scalar multiplication operation T_{mul} takes 0.54 and 0.34 ms on the device side and the infrastructure side, respectively, while the ECC addition operation T_{add} requires 0.0048 ms and 0.0026 ms. The comparative results of computational overhead for each scheme are presented in Table I.

TABLE I
COMPUTATIONAL COST COMPARISON

Scheme	Device	Infrastructure(single)	Infrastructure (batch)
[9]	$4T_{mul} + T_{add}$	$4T_{mul} + T_{add}$	$4nT_{mul} + nT_{add}$
[10]	$4T_{mul}$	$4T_{mul}$	$4nT_{mul}$
[11]	$7T_{mul} + T_{add}$	$5T_{mul} + 2T_{add}$	$(2 + 3n)T_{mul} + 3nT_{add}$
[12]	$5T_{mul} + 2T_{add}$	$5T_{mul} + 3T_{add}$	$5nT_{mul} + 3nT_{add}$
Ours	$3T_{mul}$	$4T_{mul} + T_{add}$	$(1 + 3n)T_{mul} + nT_{add}$

Fig.2 displays computational cost comparisons during individual authentication procedures. The evaluation covers device side, infrastructure side, and total overhead components. Our scheme achieves substantially lower computational cost on the device side compared to the scheme [9], [10], [11], and [12]. On the infrastructure side, our scheme exhibits overhead similar to the scheme [9] and [10] but shows a notable reduction compared to the scheme [11] and [12]. In terms of total computational cost, our scheme reaches the minimum value. The improvement reaches approximately 15.27% compared to the second-best scheme.

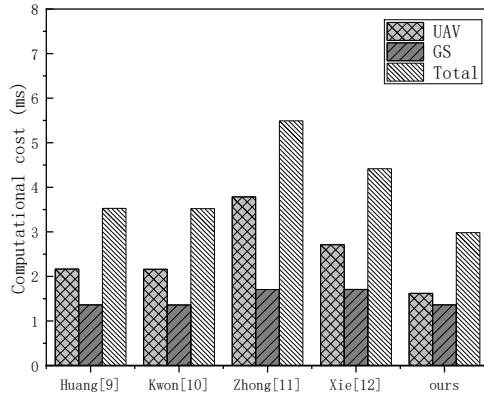


Fig. 2. Computational cost comparison for individual authentication.

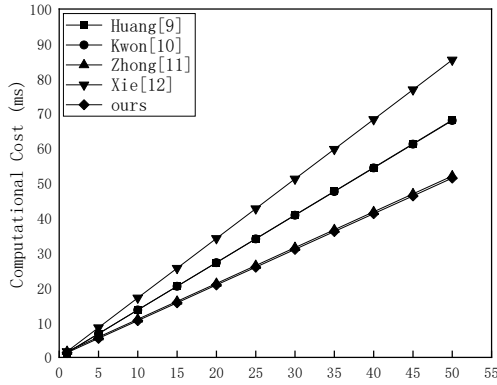


Fig. 3. Computational cost comparison for batch authentication.

Fig.3 presents the evolution of computational overhead on the infrastructure side with an increasing number of authenticated nodes n . All schemes exhibit linear growth characteristics. The scheme [12] maintains the highest computational overhead. In contrast, our scheme maintains the lowest infrastructure side overhead. This pattern verifies that our scheme maintains a reduced computational burden while guaranteeing security during large-scale UAV batch authentication, demonstrating excellent scalability.

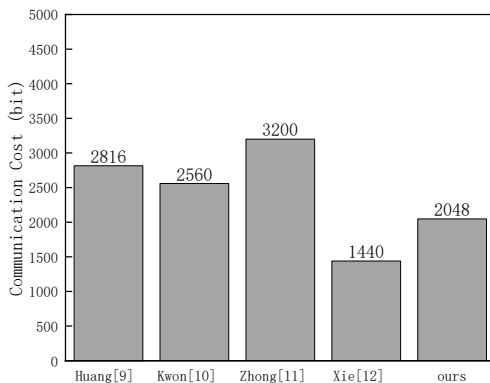


Fig. 4. Comparison of communication cost.

As depicted in Fig.4, our scheme's communication cost totals 2048 bits. In this experiment, we define the ECC length as 320 bits, the PUF challenge value length as 128 bits, the timestamp as 32 bits, and all other parameters as 256 bits. This measure slightly surpasses the scheme [12], which implements only unilateral authentication. However, it represents an approximately 36% reduction compared to the highest cost. Compared to scheme [9] and [10], it achieves approximately 27.3% and 20% cost reductions, respectively. Our scheme reduces communication overhead through an optimized message architecture while preserving security. Although the scheme [12] shows better communication efficiency, our solution exhibits a superior balance between security properties and computational performance.

For trust model validation, we simulate 100 cross-domain UAVs operating across three cycles, with 20 percent nodes designated malicious. The trust threshold is set to 0.5. We initially investigated how the robustness factor k influences false positive rates, with results shown in Fig.5 during initial cycles. At $k = 0$, the normal and malicious UAV trust distributions exhibit significant overlap. Some nodes remain below the threshold, complicating accurate identification. At $k = 1$, the strengthened penalty mechanisms still leave certain normal nodes below the threshold, sustaining a high false positive probability. At $k = 2$, all normal UAVs attain trust values above the threshold. This validates that the appropriate configuration k effectively accommodates occasional failures from environmental variations, thereby improving the robustness of the system.

In multicycle trust assessment experiments, the robustness factor k is fixed at 2. Fig.6 reveals that during cycle one, the robustness factor effectively mitigates trust fluctuations caused by sporadic failures. All node trust values stay above the threshold, avoiding false positives against normal nodes initially. In cycle two, persistent authentication failures from malicious nodes are registered in the system. Their trust values decline markedly through penalty mechanisms, and the majority of malicious nodes drop below the threshold. In cycle three, malicious behavioral nodes are fully captured. Through combined historical decay and continuous penalties, all malicious node trust values fall below the threshold, achieving complete identification. Experimental results confirm that our trust model accurately discriminates node behaviors in multicycle operations. It preserves both initial robustness and long-term operational security, establishing a reliable basis for UAV swarm security management.

VI. CONCLUSION

This paper addresses the challenges of low authentication efficiency and susceptibility to two types of pollution attacks in cross-domain batch authentication for UAV networks. A mechanism utilizing PUFs is designed for dynamic cross-domain key updates, ensuring both forward key security and resistance to physical capture attacks. Additionally, a blockchain-based trust management mechanism is developed, which enables rapid detection and isolation of legitimate UAVs that display

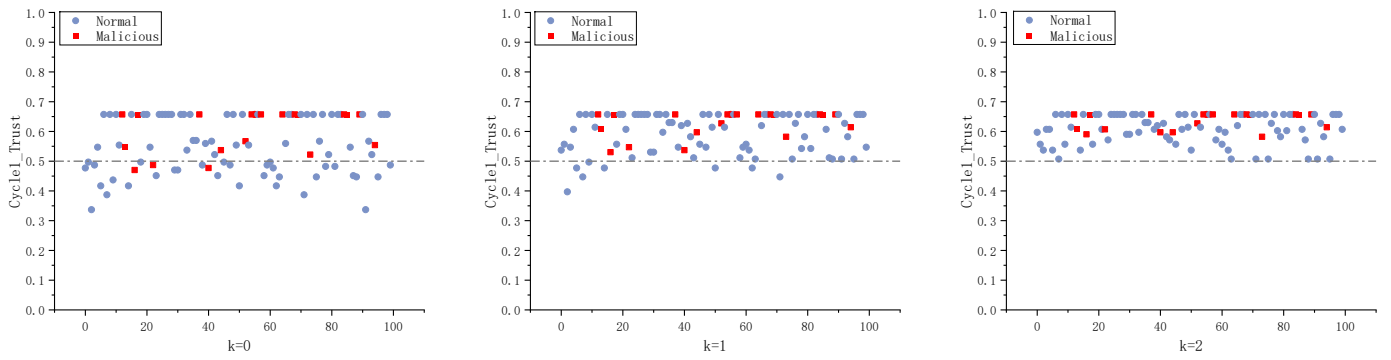


Fig. 5. Impact of the robustness factor on the false positive rate.

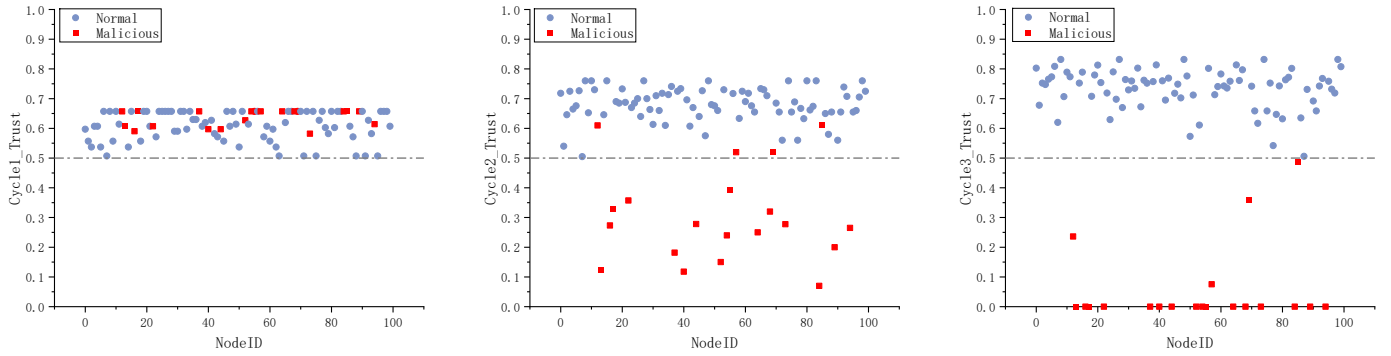


Fig. 6. Trust evaluation results over multiple cycles.

malicious behavior. Security analysis and experimental results demonstrate that the proposed scheme successfully withstands various known attacks while offering superior computational performance compared to related methods. In future research, we intend to investigate deep learning-based trust mechanisms to achieve fully intelligent cross-domain batch authentication in UAV networks.

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